

Last name \_\_\_\_\_

First name \_\_\_\_\_

**LARSON—MATH 756—CLASSROOM WORKSHEET 07**  
**Hall and König's Theorems**

**Review**

1. (Berge's Theorem) Prove: if  $G$  is a graph with matching  $M$ ,  $M$  is maximum if and only if  $G$  has no  $M$ -augmenting path. How can Berge's Theorem be used to find a non-exponential algorithm for finding a maximum matching in a *bipartite* graph?

**New**

1. What is Hall's Condition?
  
  
  
  
  
  
  
  
  
  
2. What is Hall's Theorem?
  
  
  
  
  
  
  
  
  
  
3. What is the "easy" direction of Hall's Theorem?

4. Why is the “hard” direction true? (and what is the main proof idea)?

5. What is König’s Theorem?

6. What is the “easy” direction of König’s Theorem?

7. Why is the “hard” direction true? (and what is the main proof idea)?